D3.3 Homeostasis (HL)

NAME	DATE	TOTAL
		_____ / 30

About 30 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

6 MARKS

1 **DEFINE** osmoregulation. [D3.3.7] [1]

2 **NAME** the part of the nephron where ultrafiltration takes place. [D3.3.8] [1]

3 **NAME** the gland that secretes ADH. [D3.3.10] [1]

4 **STATE** where the osmoreceptors that control ADH secretion are found. [D3.3.10] [1]

5 **STATE** two components of blood that are not found in the glomerular filtrate. [D3.3.8] [2]

Section B: Understanding

12 MARKS

6 **EXPLAIN** how the structure of the glomerulus and Bowman's capsule produces the filtrate. [D3.3.8] [3]

7 **EXPLAIN** how the cells of the proximal convoluted tubule are adapted for reabsorption. [D3.3.8] [3]

8 **EXPLAIN** the role of the loop of Henle. [D3.3.9] [3]

9 EXPLAIN how drinking a large volume of water changes the urine. [D3.3.10] [3]

Section C: Working with evidence

8 MARKS

All data in this section are invented for practice. The patterns are realistic.

10 [D3.3.8]

Samples of blood plasma, glomerular filtrate and urine were analysed in a healthy person.

CONCENTRATION / G DM^{-3}	PLASMA	FILTRATE	URINE
PROTEIN	75	0.05	0
GLUCOSE	1.0	1.0	0
UREA	0.3	0.3	20
SODIUM IONS	3.2	3.2	3.5

a EXPLAIN why the filtrate contains almost no protein. [1]

b EXPLAIN the glucose concentrations in the filtrate and in the urine. [2]

c CALCULATE how many times more concentrated urea is in urine than in plasma. [1]

d EXPLAIN why urea becomes so concentrated in urine. [2]

e PREDICT, with a reason, how the urine would differ in a person with untreated diabetes and a plasma glucose of 3 g dm^{-3}. [2]

Section D: Longer answer

4 MARKS

11 EXPLAIN how blood flow to skeletal muscle and to the gut changes during vigorous exercise. [D3.3.11] [4]